

Lecture 2 - Fermionic Quantum Error Correction

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1 Introduction

Hands-on 1: What problem can we solve with a quantum computer that will win us a Nobel prize? Answer is *room temperature superconductivity*.

1.1 Who wants to be a Nobel laureate?

The goal of this lecture will be to introduce students to the landscape of fermionic error correction by using a very practical and critical example of *superconductivity detection* inspired by this Quantinuum paper (1).

You have stumbled upon some hypothetical material which you believe will be a room temperature superconductor!!! You have also come up with properties of this material (e.g. electrons to holes ratio, doping, etc). Now your job is to prove that your hypothetical material is indeed a room temperature superconductor (RTS) under the following constraints:

1. You do not have enough funding to synthesize and grow this material from scratch and repeat multiple experiments to prove RTS.
2. You do have a quantum computer but it can only work in either of the following modes: standard qubit based (like IBM's superconducting devices) or fermionic i.e. you do not need to map your Hamiltonian to Paulis.

You need to run your simulation on actual quantum computer using techniques to either detect or correct any errors and it should be replicable across any quantum computer i.e. obtain the same results with whatever underlying error correction method is used. How will you do it?

There are two approaches: for qubit based we will discuss fermion-to-qubit mappings and how they can detect/correct errors; and for native fermionic devices we will show how Majorana fermionic error correcting codes can correct phase errors and atom loss errors. But before getting into that, let's brush up some prerequisites.

1.2 How to detect superconductivity?

From our current understanding of high temperature superconductors, Cooper pairs are essential for superconductivity (2). So we track the pairing correlating observable $p(r) = \langle c_i c_j c_k^\dagger c_l^\dagger + h.c. \rangle$, which tells what is the overlap of wavefunction of cooper pair at pair of sites (i, j) and (k, l) at distance r from each other. Lattices can be of any shape but we will consider a 2D grid here and for simplicity and all our fermions are polarized to a spin in a single direction.

1.3 Hubbard model

The Hubbard model is a minimal model describing interacting fermions (particles with half-integer spin) on a lattice (3). When nearest-neighbor interaction is included, the resulting extended Hubbard model is governed by the Hamiltonian:

$$H = -t \sum_{\langle i,j \rangle, \sigma} (\hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} + \text{h.c.}) + U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} + V \sum_{\langle i,j \rangle} \hat{n}_i \hat{n}_j \quad (1)$$

where i and j denote lattice sites; $\langle i, j \rangle$ indicates summation over nearest-neighbor pairs; $\sigma \in \{\uparrow, \downarrow\}$ indexes the spin species; $\hat{c}_{i\sigma}^\dagger$ and $\hat{c}_{i\sigma}$ are the creation and annihilation operators; $\hat{n}_{i\sigma} = \hat{c}_{i\sigma}^\dagger \hat{c}_{i\sigma}$ is the spin-resolved number operator; $\hat{n}_i = \hat{n}_{i\uparrow} + \hat{n}_{i\downarrow}$ is the total number of fermions at site i ; and h.c. denotes the Hermitian conjugate. Here, t represents the hopping amplitude, U the on-site Coulomb repulsion, and V the inter-site Coulomb repulsion. Despite its simplicity, this model captures essential electron correlation physics driven by local interactions within a single orbital (4).

2 Run simulation on error-corrected qubit hardware

Steps required to simulate Hamiltonian on qubits:

1. Prepare initial state (here Gibb's state is used since we want to observe temperature dependence, refer to (5; 6) for more details);
2. Map your Hamiltonian to Paulis. We will go over how to do this using two methods (Jordan-Wigner and Bravyi-Kitaev-SuperFast) in this section.
3. Trotter evolve the initial state under your Hamiltonian i.e : if $H = \sum_k H_k$ s.t. $\exists(i, j)$ for which $[H_i, H_j] \neq 0$ then $e^{-iHt}|\psi\rangle \approx (\prod_k e^{\frac{-iH_k t}{n}})^n |\psi\rangle$ is Trotter evolution with n Trotter steps.
4. Measure your observable (in our case we measure $p(r) = \langle c_i c_j c_k^\dagger c_l^\dagger + \text{h.c.} \rangle$ over distance r)

2.1 Jordan-Wigner transformation

Recap on Jordan-Wigner (JW) encodings:

$$\begin{aligned} n_j &= \frac{Z_j + I}{2} \\ c_j &= \left(\prod_{k < j} Z_k \right) \frac{X_j + iY_j}{2} \\ c_j^\dagger &= \left(\prod_{k < j} Z_k \right) \frac{X_j - iY_j}{2} \end{aligned} \quad (2)$$

Hands-on 2: Show how JW in a snake pattern can be used for simulating 2D lattice with the 4x4 lattice shown in Fig. 1. Now pick any vertical edge and workout how $c_i^\dagger c_j$ maps to Paulis, let's say, 8 – 12

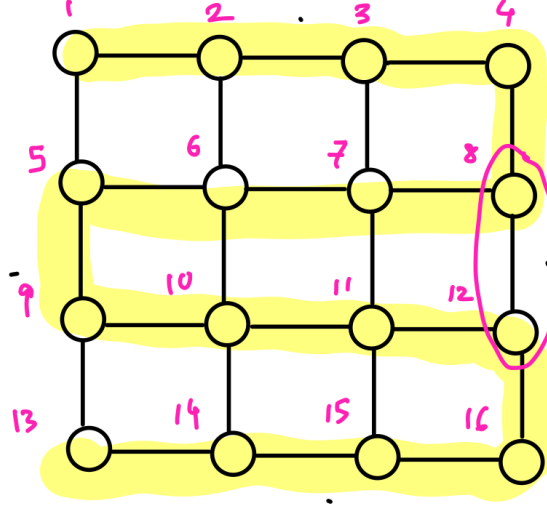


Figure 1: Highlighted in yellow is the Jordan-Wigner encoding used to map 4×4 grid of fermions onto a 4×4 grid of qubits.

$$\begin{aligned}
c_8^\dagger &= Z_1 \dots Z_7 \frac{X_8 - iY_8}{2} \\
c_{12} &= Z_1 \dots Z_{11} \frac{X_{12} + iY_{12}}{2} \\
\therefore c_8^\dagger c_{12} &= \left(\frac{X_8 Z_8 - iY_8 Z_8}{2} \right) Z_9 Z_{10} Z_{11} \left(\frac{X_{12} + iY_{12}}{2} \right) \\
\text{and } n_8 n_{12} &= \frac{1}{4} (I + Z_8 + Z_{12} + Z_8 Z_{12})
\end{aligned} \tag{3}$$

This shows how the non locality is manifested in operators, i.e. one local interaction turned into weight 5 Pauli operator. Mapping number operators n will not create such long non-local Z strings shown in Eq. (3). Now to perform even one Trotter step evolution of the Hubbard model for our 4×4 grid of qubits, we have 48 terms (24 edge interaction terms leading to 48 terms counting hermitian conjugates: $c_i^\dagger c_j + c_j^\dagger c_i$) of high weight Paulis.

Hands-on 3: "Why can't we simply run the simulation with Jordan-Wigner (JW) transformation on logical qubits within a surface or Steane code using Trotterization?". To execute this, high-weight, non-local Pauli strings must first be decomposed into Clifford+ T gates. Because fault tolerant T gates require costly magic state distillation, the gate overhead scales drastically even for a single Trotter step comprising of non-commuting, high-weight Hamiltonian terms. Consequently, standard JW mapping combined with naive Trotterization is practically infeasible for fault-tolerant fermionic simulation.

2.2 Bravyi-Kitaev SuperFast (BKSF) encoding

In this section, we will borrow the ideas of quantum error correction to use multiple physical qubit to encode one logical fermion which provides a way to detect any single-qubit errors occurring during simulation. The Jordan-Wigner mapping cannot natively detect single-qubit

errors because it is a bijection between the N -mode fermionic Fock space and the N -qubit Hilbert space. Because the logical fermionic state space completely fills the 2^N -dimensional qubit space, any qubit error transforms the state into another valid computational basis state rather than projecting it into non-physical error-detecting subspace.

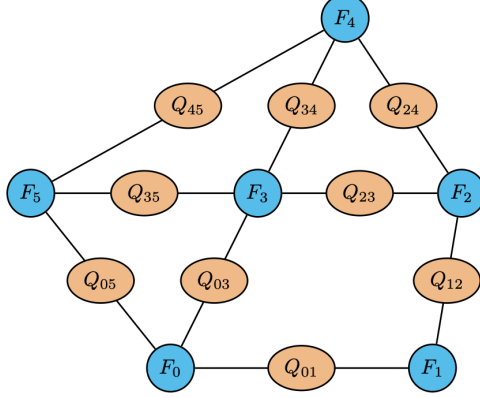


Figure 2: BKSF (Bravyi-Kitaev SuperFast encoding) on a graph (7): the vertices (blue circles) of the graph represent fermionic modes; fermions can hop between neighboring vertices. One qubit (orange ellipses) is introduced for each edge of the graph. The fermionic state is mapped to a subspace of the qubits.

Bravyi and Kitaev (8) considered fermionic hopping problem defined on a graphs G , where fermions can hop between neighboring vertices. One qubit is introduced for each edge of the graph G shown in Fig. 2, and the fermionic state is encoded in a subspace of the qubits. The goal is to find qubit operators that share the same algebra as the fermionic operators.

2.2.1 Recap on Majorana fermions

The single mode Majorana operators are defined as linear combinations of the fermionic ladder operators

$$\begin{aligned}\gamma_k &= c_k^\dagger + c_k \\ \tilde{\gamma}_k &= i(c_k^\dagger - c_k)\end{aligned}\tag{4}$$

where $c_k^\dagger$ and c_k are the fermionic creation and annihilation operators, respectively. Fermionic annihilation and creation operators can also be represented in terms of Majorana operators:

$$\begin{aligned}c_k &= \frac{1}{2}(\gamma_k + i\tilde{\gamma}_k) \\ c_k^\dagger &= \frac{1}{2}(\gamma_k - i\tilde{\gamma}_k)\end{aligned}\tag{5}$$

Majorana operators are Hermitian ($\gamma_k^\dagger = \gamma_k$, $\tilde{\gamma}_k^\dagger = \tilde{\gamma}_k$) and satisfy the following relations:

$$\begin{aligned}\{\gamma_k, \gamma_j\} &= \{\tilde{\gamma}_k, \tilde{\gamma}_j\} = 2\delta_{jk} \\ \{\gamma_k, \tilde{\gamma}_j\} &= 0\end{aligned}\tag{6}$$

where $\{\cdot, \cdot\}$ is the anticommutator and δ the Kronecker delta function.

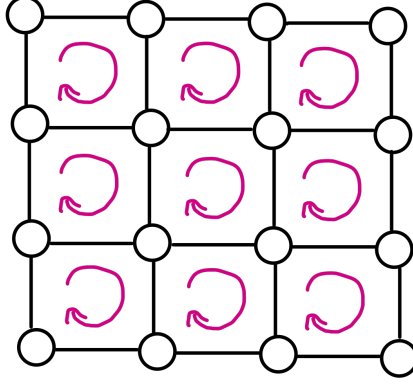


Figure 3: Elementary loops on a 2D square lattice. The independent closed paths are generated by fundamental 4-site square plaquettes $p = (k_0, k_1, k_2, k_3)$ of length $l = 4$, where each plaquette defines a local conserved operator $\mathcal{B}_p = \mathcal{B}_{k_0 k_1} \mathcal{B}_{k_1 k_2} \mathcal{B}_{k_2 k_3} \mathcal{B}_{k_3 k_0} = \mathbb{I}$.

2.2.2 Redefine fermionic operators using Majorana edge and vertex operators

Now that we have revisited Majorana fermion operators, let's see what happens when we rewrite hopping term ($c_j^\dagger c_k + h.c.$) and number operator ($c_k^\dagger c_k$) in Hubbard model from Eq. (1) in terms of Majorana operators.

For the number operator:

$$\begin{aligned} n_k &= c_k^\dagger c_k = \frac{1}{4}(\gamma_k - i\tilde{\gamma}_k)(\gamma_k + i\tilde{\gamma}_k) \\ &= \frac{1}{4}(\gamma_k^2 + \tilde{\gamma}_k^2 + i(\gamma_k \tilde{\gamma}_k - \tilde{\gamma}_k \gamma_k)) \end{aligned}$$

substituting the anticommutation relations from Eq. (6), $\gamma_k^2 = \tilde{\gamma}_k^2 = \mathbb{I}$ and $\tilde{\gamma}_k \gamma_k = -\gamma_k \tilde{\gamma}_k$ above, we get:

$$n_k = \frac{1}{2}(\mathbb{I} + i\gamma_k \tilde{\gamma}_k) \quad (7)$$

For the hopping term:

$$\begin{aligned} c_j^\dagger c_k + c_k^\dagger c_j &= \frac{1}{4}((\gamma_j - i\tilde{\gamma}_j)(\gamma_k + i\tilde{\gamma}_k) + (\gamma_k - i\tilde{\gamma}_k)(\gamma_j + i\tilde{\gamma}_j)) \\ &= \frac{1}{4}((\gamma_j \gamma_k + \gamma_k \gamma_j) + (\tilde{\gamma}_j \tilde{\gamma}_k + \tilde{\gamma}_k \tilde{\gamma}_j) + i(\gamma_j \tilde{\gamma}_k - \tilde{\gamma}_k \gamma_j) - i(\tilde{\gamma}_j \gamma_k - \gamma_k \tilde{\gamma}_j)) \end{aligned}$$

substituting the anticommutation relations from Eq. (6), $(\gamma_j \gamma_k + \gamma_k \gamma_j) = (\tilde{\gamma}_j \tilde{\gamma}_k + \tilde{\gamma}_k \tilde{\gamma}_j) = 0$, $\tilde{\gamma}_k \gamma_j = -\gamma_j \tilde{\gamma}_k$ and $\gamma_k \tilde{\gamma}_j = -\tilde{\gamma}_j \gamma_k$ above, we get:

$$\begin{aligned} c_j^\dagger c_k + c_k^\dagger c_j &= \frac{i}{2}(\gamma_j \tilde{\gamma}_k - \tilde{\gamma}_j \gamma_k) \\ &= \frac{i}{2}(\gamma_j \gamma_k \gamma_k \tilde{\gamma}_k - \tilde{\gamma}_j \gamma_j \gamma_j \gamma_k) = \frac{i}{2}(\gamma_j \gamma_k \gamma_k \tilde{\gamma}_k + \gamma_j \tilde{\gamma}_j \gamma_j \gamma_k) \\ &= -\frac{i}{2}((i\gamma_j \gamma_k)(i\gamma_k \tilde{\gamma}_k) + (i\gamma_j \tilde{\gamma}_j)(i\gamma_j \gamma_k)) \end{aligned} \quad (8)$$

Finally we introduce edge and vertex operators. For a lattice of where vertices $\{V\}$ are fermions and edges $\{E\}$ represent interaction between neighboring fermions, we define

- Vertex operator: $\mathcal{A}_k = i\gamma_k \tilde{\gamma}_k, \forall k \in V$

- Edge operator: $\mathcal{B}_{jk} = i\gamma_j\gamma_k, \forall (j, k) \in E$

Not surprisingly, Eq. (7) and Eq. (8) can be written in terms of the vertex operators $\mathcal{A}_k = i\gamma_k\tilde{\gamma}_k$ and edge operators $\mathcal{B}_{jk} = i\gamma_j\gamma_k$:

$$\begin{aligned} n_k &= \frac{1}{2}(\mathbb{I} + \mathcal{A}_k) \\ c_j^\dagger c_k + c_k^\dagger c_j &= -\frac{i}{2}(\mathcal{B}_{jk}\mathcal{A}_k + \mathcal{A}_j\mathcal{B}_{jk}) \end{aligned} \quad (9)$$

The edge and vertex operators satisfy the following commutation relationships that preserve the underlying fermionic algebra:

$$\begin{aligned} \mathcal{A}_j\mathcal{A}_k &= \mathcal{A}_k\mathcal{A}_j \\ \mathcal{A}_l\mathcal{B}_{jk} &= (-1)^{\delta_{jl}+\delta_{kl}}\mathcal{B}_{jk}\mathcal{A}_l \\ \mathcal{B}_{lm}\mathcal{B}_{jk} &= (-1)^{\delta_{jl}+\delta_{kl}+\delta_{jm}+\delta_{km}}\mathcal{B}_{jk}\mathcal{B}_{lm} \end{aligned} \quad (10)$$

Hands-On 4: Show that $(-i)^l \mathcal{B}_{k_0 k_1} \mathcal{B}_{k_1 k_2} \dots \mathcal{B}_{k_{l-1} k_0} = \mathbb{I}$ over any closed loop of length l is a conserved quantity.

$$\begin{aligned} \mathcal{B}_{k_0 k_1} \mathcal{B}_{k_1 k_2} \dots \mathcal{B}_{k_{l-1} k_0} &= (i\gamma_{k_0}\gamma_{k_1})(i\gamma_{k_1}\gamma_{k_2}) \dots (i\gamma_{k_{l-1}}\gamma_{k_0}) \\ &= i^l \gamma_{k_0}(\gamma_{k_1}\gamma_{k_1})(\gamma_{k_2}\gamma_{k_2}) \dots (\gamma_{k_{l-1}}\gamma_{k_{l-1}})\gamma_{k_0} \end{aligned}$$

Since $\gamma_k^2 = \mathbb{I}$ and $k_l = k_0$ for a closed loop

$$\mathcal{B}_{k_0 k_1} \mathcal{B}_{k_1 k_2} \dots \mathcal{B}_{k_{l-1} k_0} = i^l \mathbb{I} \quad (11)$$

Multiplying both sides by $i^{-l} = (-i)^l$:

$$(-i)^l \mathcal{B}_{k_0 k_1} \mathcal{B}_{k_1 k_2} \dots \mathcal{B}_{k_{l-1} k_0} = \mathbb{I}$$

Because this loop operator is proportional to the identity operator $\mathbb{I}$, it trivially commutes with any Hamiltonian ($[\mathcal{B}_{\text{loop}}, H] = 0$), making it a conserved quantity. The total number of independent closed paths for a connected lattice is $N_E - N_V + 1$, where N_E (N_V) is the total number of edges (vertices). The independent closed loops on a 2D degree-4 lattice are generated by elementary square plaquettes, as shown in Fig. 3.

2.2.3 Map BKSF to Pauli operators

To map $\mathcal{A}_k$ and $\mathcal{B}_{jk}$ to Pauli, we first choose an arbitrary ordering of the edges for each vertex of G , e.g. the counterclockwise order shown in Fig. 4 $(f, h) < (f, i) < (f, g) < (f, j)$. This order is not unique, and the edges can be chosen in any order but the same order must be used for all the vertices.

The vertex operators are mapped to the Pauli as follows:

$$\tilde{A}_k = \prod_{j:(j,k) \in E} Z_{jk} \quad (12)$$

and the edge operator are mapped as:

$$\tilde{B}_{jk} = \epsilon_{jk} X_{jk} \prod_{l:(j,l) < (j,k)} Z_{jl} \prod_{m:(k,m) < (k,j)} Z_{km} \quad (13)$$

where $\sim$ in $\tilde{A}_k$ and $\tilde{B}_{jk}$ represents that these are qubit i.e. logical fermionic operator, the antisymmetric coefficients $\epsilon_{jk} = -\epsilon_{kj} = \pm 1$ can be chosen arbitrarily for each edge (j, k) . The

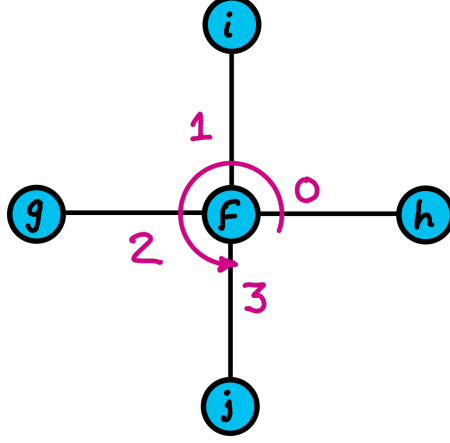


Figure 4: Counterclockwise order of edges chosen for BKSF mapping.

qubit operators $\tilde{A}_k$ and $\tilde{B}_{jk}$ satisfy the same commutation relations as those of the fermionic operators in Eq. (10).

The edge operator mapping in Eq. (13) is a bit intimidating to look at but let's do an example for better illustration. Consider the 3×3 grid of fermions shows in Fig. 5 for which we will derive the edge operators in terms of Pauli for the plaquette (m, n, q, p, m) following the same counterclockwise edge ordering from Fig. 4.

The vertex operators are fairly simple to map to Pauli:

$$\begin{aligned}\tilde{\mathcal{A}}_m &= Z_{mn}Z_{mj}Z_{ml}Z_{mp} \\ \tilde{\mathcal{A}}_n &= Z_{nk}Z_{nm}Z_{nq} \\ \tilde{\mathcal{A}}_q &= Z_{qn}Z_{qp} \\ \tilde{\mathcal{A}}_m &= Z_{pq}Z_{pm}Z_{po}\end{aligned}$$

Now, for the edge operators:

1. (m, n) : label of edge (m, n) at vertex m is 0, edges containing m are $\{(m, j), (m, l), (m, p)\}$ with labels $\{1, 2, 3\}$, respectively. Since no edges containing m have a label strictly smaller than 0, no Z operators originate from vertex m . Similarly, label of edge (m, n) at vertex n is 2, edges containing n are $\{(n, k), (n, q)\}$ with labels $\{1, 3\}$, respectively. Edge (n, k) has label $1 < 2$, meaning it precedes edge (m, n) and requires a Z operator on edge qubit (n, k) . Thus, logical edge operators is:

$$\tilde{\mathcal{B}}_{mn} = X_{mn}Z_{nk}$$

2. (n, q) : Edges preceding (n, q) are $\{(n, k), (n, m)\}$. No edges preceding (q, n) .

$$\tilde{\mathcal{B}}_{nq} = X_{nq}Z_{nk}Z_{nm}$$

3. (q, p) : Edges preceding (q, p) are $\{(q, n)\}$. No edges preceding (p, q) .

$$\tilde{\mathcal{B}}_{qp} = X_{qp}Z_{qn}$$

4. (p, m) : Edges preceding (p, m) are $\{(p, q)\}$. Edges preceding (m, p) are $\{(m, n), (m, j), (m, l)\}$.

$$\tilde{\mathcal{B}}_{pm} = X_{pm}Z_{pq}Z_{mn}Z_{mj}Z_{ml}$$

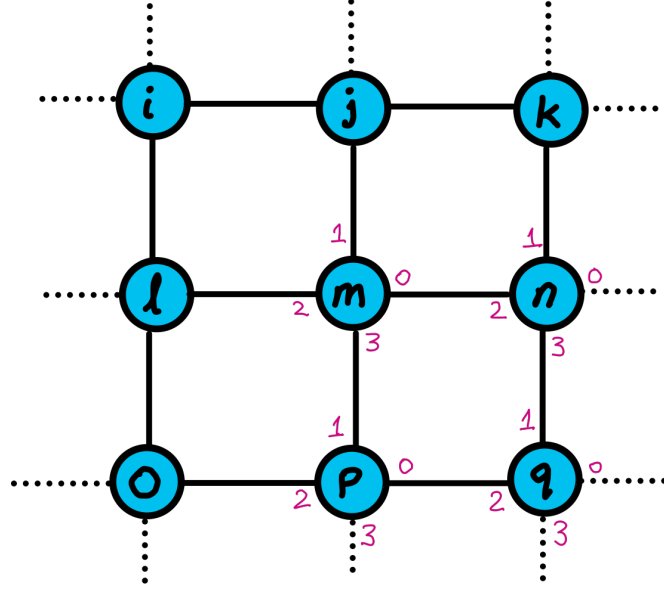


Figure 5: BKSF on a 2D lattice with a specific ordering of the edges. The dotted edges are not real and are only drawn to show the edge ordering. This example is adapted from (7)

Hands-on 5: Derive stabilizer using the conserved quantity in Eq. (11) for plaquette (m, n, q, p, m) in terms of Pauli operators and reason why BKSF is a $d = 2$ code.

Let S_{mnqpm} be the stabilizer for plaquette (m, n, q, p, m) . Using Eq. (11)

$$\begin{aligned}
 S_{mnqpm} &= (-i)^l \tilde{\mathcal{B}}_{mn} \tilde{\mathcal{B}}_{nq} \tilde{\mathcal{B}}_{qp} \tilde{\mathcal{B}}_{pm} \\
 &= (-i)^4 (X_{mn} \textcolor{red}{Z}_{nk}) (X_{nq} \textcolor{red}{Z}_{nk} \textcolor{red}{Z}_{nm}) (X_{qp} Z_{qn}) (X_{pm} Z_{pq} \textcolor{red}{Z}_{mn} Z_{mj} Z_{ml}) \\
 &= (X_{mn}) (X_{nq}) (X_{qp} Z_{qn}) (X_{pm} Z_{pq} Z_{mj} Z_{ml}) \\
 &= (X_{mn}) (X_{nq} Z_{qn}) (-X_{qp} Z_{pq}) (-X_{pm}) (Z_{mj} Z_{ml}) \\
 &= X_{mn} Y_{nq} Y_{pq} X_{pm} Z_{mj} Z_{ml}
 \end{aligned} \tag{14}$$

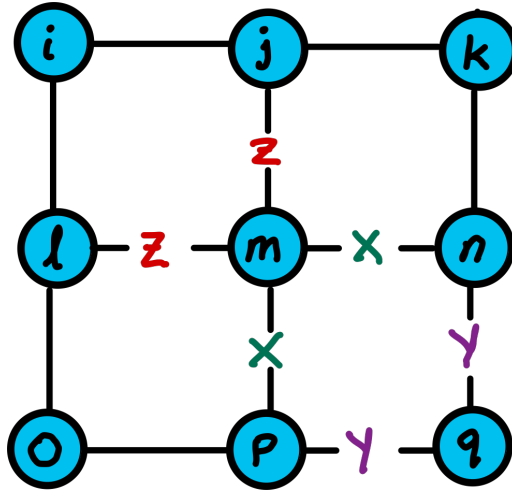


Figure 6: Plaquette (m, n, q, p, m) stabilizer for the BKSF code given the vertex and edge ordering in Fig. 4

Looking at Fig. 6, a single Z error on edge (n, q) anti-commutes with Y_{nq} generating the exact same syndrome signature as a single Z error on the neighboring edge (p, q) . Moreover, a single X error anti-commutes with Y on either of the edge qubits (n, q) or (p, q) , thus generating the exact same syndrome signature as a single Z error. Because these distinct single-qubit errors produce identical syndromes, no decoder can distinguish between them. Consequently, misidentifying the faulty edge leaves an uncorrected, non-trivial weight-2 Pauli operator ($Z_{nq}Z_{pq}$) or ($X_{nq}X_{pq}$) in the code space, thereby fixing the code distance at $d = 2$.

Unlike Jordan-Wigner mapping, where operator weight scales with system size N , BKSF mapping yields Pauli operator weights bounded by $O(d)$, where d is the maximum degree of the lattice. Because this weight remains constant as the lattice expands, the representation enables superfast simulation. We can now run our Trotterized RTS simulation on qubits and have stabilizer check cycles which can detect error and post-select the results based on syndrome measurements.

2.3 Additional fermion to qubit encoding with error correcting abilities

Recent developments in fermion-to-qubit encodings have significantly expanded beyond the distance 2 limits of standard BKSF by introducing diverse mechanisms to achieve fault-tolerant error correction. While near-term encodings prioritize minimal qubit overhead such as super-compact encodings operating at 1.25 qubits per fermion without error correction (9), others actively embed error protection into the local lattice structure. Frameworks like the Majorana Loop Stabilizer Code (MLSC) (7) and Generalized Superfast Encodings (GSE) (10) utilize loop stabilizers or higher-degree interaction graphs ($d \geq 6$) to resolve phase-error degeneracies, enabling full single-qubit error correction ($d = 3$). Moving toward fault-tolerant regimes, modern architectures decouple code distance from local graph degree altogether: recent constructions leverage topological defects in 2D surface codes to simulate fermions with arbitrary code distance (11), or concatenate low-overhead local mappings directly with high-distance 2D and 3D Majorana fermionic color codes (12).

3 Run simulation on a fermionic device

We assume access to a digital fermionic quantum computer capable of running a universal fermionic gate set. The steps to run our RTS simulation experiment on a fermionic device remain the same as discussed in Section 2, except for the Pauli mapping step. Because we are using a digital fermionic device, we still need to Trotterize our natively fermionic Hamiltonian.

3.1 Majorana repetition code

Some of this subsection will be revision of Section 3.4.3 from Sam’s lecture notes. Phase errors represent the dominant source of noise in standard fermionic platforms, such as neutral atom arrays (13). A phase error at site k is characterized by the local occupation operator n_k . To mitigate this, a Majorana repetition code can be employed to detect and correct arbitrary phase errors.

Following Kitaev’s 1D chain model (14), a logical fermionic mode can be distributed across m physical fermions. While any complex fermionic mode can be algebraically decomposed into a pair of Majorana operators, localized physical Majorana zero modes (MZMs) emerge only within topologically non-trivial phases.

Let’s look at Fig. 7. Each physical fermion splits and carries two Majoranas and stabilizers pair them across neighbouring sites, so every Majorana pairs up except the two at the ends of the chain. The two edge Majoranas together make one complete logical fermionic mode. $m - 1$

stabilizers are needed for if we encode one logical fermion using m physical fermions

$$S_k = i\tilde{\gamma}_k\gamma_{k+1}, \quad k = 1, 2, \dots, m-1. \quad (15)$$



Figure 7: Stabilizer checks for Majorana repetition code pair the bulk across sites

The logical creation and annihilation operators are given by:

$$\tilde{c}^\dagger = \frac{1}{2}(\gamma_1 + i\tilde{\gamma}_m), \quad \tilde{c} = \frac{1}{2}(\gamma_1 - i\tilde{\gamma}_m) \quad (16)$$

The logical vacuum $|0_L\rangle$ and occupied state $|1_L\rangle$ span the 2-dimensional logical code space. Both states are simultaneous $+1$ eigenstates of all $m-1$ stabilizer checks, ensuring local bulk errors remain detectable:

$$S_k|0_L\rangle = +1|0_L\rangle, \quad S_k|1_L\rangle = +1|1_L\rangle \quad \forall k \in 1, \dots, m-1$$

The logical vacuum $|0_L\rangle$ is uniquely fixed by the stabilization condition alongside the logical annihilation requirement $\tilde{c}|0_L\rangle = 0$. The logical excited state is then constructed by populating the non-local mode: $|1_L\rangle = \tilde{c}^\dagger|0_L\rangle$

Hands-on 6: Show that Majorana repetition code can detect and correct any single phase error and detect (but not correct) single atom loss errors for all sites except the edge fermions.

From Eq. (7), the phase error operator at site k is $n_k = \frac{1}{2}(\mathbb{I} + i\gamma_k\tilde{\gamma}_k)$. Since the identity operator $\mathbb{I}$ commutes with all stabilizers and contributes no syndrome flips, the non-trivial error action is entirely captured by $E_k \sim i\gamma_k\tilde{\gamma}_k$. Let's compute the anti-commutator of this error with the stabilizer $S_j = i\tilde{\gamma}_j\gamma_{j+1}$:

$$\begin{aligned} \{E_k, S_j\} &= \{i\gamma_k\tilde{\gamma}_k, i\tilde{\gamma}_j\gamma_{j+1}\} \\ &= -(\gamma_k\tilde{\gamma}_k\tilde{\gamma}_j\gamma_{j+1} + \tilde{\gamma}_j\gamma_{j+1}\gamma_k\tilde{\gamma}_k) \end{aligned}$$

For site $j = k$:

$$\begin{aligned} \{E_k, S_k\} &= -(\gamma_k\tilde{\gamma}_k\tilde{\gamma}_k\gamma_{k+1} + \tilde{\gamma}_k\gamma_{k+1}\gamma_k\tilde{\gamma}_k) \\ &= -(\gamma_k\gamma_{k+1} + (-1)^2\tilde{\gamma}_k\tilde{\gamma}_k\gamma_{k+1}\gamma_k) \\ &= -(\gamma_k\gamma_{k+1} + \gamma_{k+1}\gamma_k) \\ &= -\{\gamma_k, \gamma_{k+1}\} \\ &= 0 \end{aligned} \quad (17)$$

Similarly, for site $j = k-1$:

$$\begin{aligned} \{E_k, S_{k-1}\} &= -(\gamma_k\tilde{\gamma}_k\tilde{\gamma}_{k-1}\gamma_k + \tilde{\gamma}_{k-1}\gamma_k\gamma_k\tilde{\gamma}_k) \\ &= -((-1)^2\gamma_k\gamma_k\tilde{\gamma}_k\tilde{\gamma}_{k-1} + \tilde{\gamma}_{k-1}\tilde{\gamma}_k) \\ &= -(\tilde{\gamma}_k\tilde{\gamma}_{k-1} + \tilde{\gamma}_{k-1}\tilde{\gamma}_k) \\ &= -\{\tilde{\gamma}_k, \tilde{\gamma}_{k-1}\} \\ &= 0 \end{aligned} \quad (18)$$

Hence, from Eq. (17) and Eq. (18), syndromes of stabilizers $\{S_k, S_{k-1}\}$ for a phase error at site k will be measured as $\{-1, -1\}$. The phase error $E_k = i\gamma_k\tilde{\gamma}_k$ is self-inverse ($E_k^2 = \mathbb{I}$) and serves as its own recovery operator, applying a local Z_k phase rotation (remember from Eq. (7) $i\gamma_k\tilde{\gamma}_k$ already looks like a Z_k) to return the system to the logical codespace.

Now let's see what happens if an atom loss error occurs. From Eq. (5), atom loss at site k is $c_k = \frac{1}{2}(\gamma_k + i\tilde{\gamma}_k)$, non-unitary action of which is projected by stabilizer measurement into one of two single-defect syndrome branches corresponding to γ_k and $\tilde{\gamma}_k$. If γ_k acts:

$$\begin{aligned}\{\gamma_k, S_{k-1}\} &= \{\gamma_k, i\tilde{\gamma}_{k-1}\gamma_k\} \\ &= i(\gamma_k\tilde{\gamma}_{k-1}\gamma_k + \tilde{\gamma}_{k-1}\gamma_k\gamma_k) \\ &= i(-\gamma_k\gamma_k\tilde{\gamma}_{k-1} + \tilde{\gamma}_{k-1}) \\ \{\gamma_k, S_{k-1}\} &= 0\end{aligned}$$

γ_k does not appear in $S_k = i\tilde{\gamma}_k\gamma_{k+1}$ so it commutes i.e. $[\gamma_k, S_k] = 0$. Therefore, action of γ_k leads to syndrome measurement of $\{-1, +1\}$ when stabilizers $\{S_{k-1}, S_k\}$ are measured. Similarly, action of $\tilde{\gamma}_k$ produces syndrome measurement of $\{+1, -1\}$ for stabilizers $\{S_{k-1}, S_k\}$.

We have now shown that atom loss at site k will produce syndrome of either $\{-1, +1\}$ or $\{+1, -1\}$ for stabilizers $\{S_{k-1}, S_k\}$. Consider two scenarios of atoms loss at site $k-1$ and k to see what syndromes for stabilizers $\{S_{k-2}, S_{k-1}, S_k\}$ will look like in the table below:

Atom Loss Site	$\{S_{k-2}, S_{k-1}, S_k\}$ syndrome
$k-1$	$\{-1, +1, +1\}$ or $\{+1, -1, +1\}$
k	$\{+1, -1, +1\}$ or $\{+1, +1, -1\}$

Notice how two different atom loss errors can non-deterministically produce same syndrome measurements, thus the decoder will not know where the actual error occurred. Moreover, atom loss at site $k=1$ and $k=m$ have γ_1 and $\tilde{\gamma}_m$ in them, respectively, which do not appear in any of the stabilizers, thus atom loss at edge sites can go undetected.

For our RTS simulation experiment, the Majorana repetition code is insufficient as it fails to mitigate atom loss. More critically, it is not fault-tolerant under fermionic gate operations, which transform single-phase errors into pairs of undetectable, correlated atom-loss errors (15).

3.2 Fault-Tolerant Fermionic Steane Code

In the context of encoding qubits in Majorana nanowire platforms, Bravyi, Terhal, and Leemhuis (16) noted that a particular class of qubit error correcting codes, so-called weakly self-dual Calderbank-Shor-Steane (CSS) codes, can be converted into a Majorana stabilizer code by replacing qubit operators in the stabilizers by Majorana fermions (15).

Hands-on 7: Show that fermionic Steane code obtained by replacing Z and X in qubit Steane code by γ and $\tilde{\gamma}$, respectively, can detect and correct any single-site phase and atom loss error.

Replacing the Z and X in qubit Steane code by γ and $\tilde{\gamma}$ gives us the following set of stabilizers:

$S_1 = \gamma_1\gamma_2\gamma_3\gamma_4$	$\tilde{S}_1 = \tilde{\gamma}_1\tilde{\gamma}_2\tilde{\gamma}_3\tilde{\gamma}_4$
$S_2 = \gamma_1\gamma_2\gamma_5\gamma_6$	$\tilde{S}_2 = \tilde{\gamma}_1\tilde{\gamma}_2\tilde{\gamma}_5\tilde{\gamma}_6$
$S_3 = \gamma_1\gamma_3\gamma_5\gamma_7$	$\tilde{S}_3 = \tilde{\gamma}_1\tilde{\gamma}_3\tilde{\gamma}_5\tilde{\gamma}_7$

Let's see what happens when phase error occurs on fermion 1 i.e. $E_1 = i\gamma_1\tilde{\gamma}_1$

$$\begin{aligned}\{E_1, S_1\} &= i(\gamma_1\tilde{\gamma}_1\gamma_1\gamma_2\gamma_3\gamma_4 + \gamma_1\gamma_2\gamma_3\gamma_4\gamma_1\tilde{\gamma}_1) \\ &= i(-\gamma_1\gamma_1\tilde{\gamma}_1\gamma_2\gamma_3\gamma_4 + (-1)^6\gamma_1\gamma_1\tilde{\gamma}_1\gamma_2\gamma_3\gamma_4) \\ \{E_1, S_1\} &= 0\end{aligned}$$

$$\begin{aligned}\{E_1, \tilde{S}_1\} &= i(\gamma_1\tilde{\gamma}_1\tilde{\gamma}_1\tilde{\gamma}_2\tilde{\gamma}_3\tilde{\gamma}_4 + \tilde{\gamma}_1\tilde{\gamma}_2\tilde{\gamma}_3\tilde{\gamma}_4\gamma_1\tilde{\gamma}_1) \\ &= i(\gamma_1\tilde{\gamma}_2\tilde{\gamma}_3\tilde{\gamma}_4 + (-1)^6\tilde{\gamma}_1\gamma_1\tilde{\gamma}_1\tilde{\gamma}_2\tilde{\gamma}_3\tilde{\gamma}_4) \\ &= i(\gamma_1\tilde{\gamma}_2\tilde{\gamma}_3\tilde{\gamma}_4 - \tilde{\gamma}_1\tilde{\gamma}_1\gamma_1\tilde{\gamma}_2\tilde{\gamma}_3\tilde{\gamma}_4) \\ \{E_1, \tilde{S}_1\} &= 0\end{aligned}$$

Similarly, E_1 will anti-commute with all the stabilizers and produce syndrome of $\{-1, -1, -1, -1, -1, -1\}$ for stabilizers $\{S_1, S_2, S_3, \tilde{S}_1, \tilde{S}_2, \tilde{S}_3\}$, and the corresponding correction of $Z_1 \sim i\gamma_1\tilde{\gamma}_1$ can be applied. Table below shows what syndromes are produced for any phase error:

Phase error site	Syndrome ($S_1, \tilde{S}_1$)	Syndrome ($S_2, \tilde{S}_2$)	Syndrome ($S_3, \tilde{S}_3$)
1	-1, -1	-1, -1	-1, -1
2	-1, -1	-1, -1	+1, +1
3	-1, -1	+1, +1	-1, -1
4	-1, -1	+1, +1	+1, +1
5	+1, +1	-1, -1	-1, -1
6	+1, +1	-1, -1	+1, +1
7	+1, +1	+1, +1	-1, -1

Now let's look at an atom loss error at site $k = 1$, $c_1 = \frac{1}{2}(\gamma_1 + i\tilde{\gamma}_1)$. Similar to what we discussed earlier, the non-unitary action of c_1 is projected by stabilizer measurements into one of two single-defect syndrome branches corresponding to γ_1 or $\tilde{\gamma}_1$.

$$\begin{aligned}\{\gamma_1, S_1\} &= \{\gamma_1, \gamma_1\gamma_2\gamma_3\gamma_4\} \\ &= \gamma_1\gamma_1\gamma_2\gamma_3\gamma_4 + \gamma_1\gamma_2\gamma_3\gamma_4\gamma_1 \\ &= \gamma_2\gamma_3\gamma_4 + (-1)^3\gamma_1\gamma_1\gamma_2\gamma_3\gamma_4 \\ \{\gamma_1, S_1\} &= 0\end{aligned}$$

γ_1 does not appear in $\tilde{S}_1 = \tilde{\gamma}_1\tilde{\gamma}_2\tilde{\gamma}_3\tilde{\gamma}_4$ so it commutes i.e. $[\gamma_1, \tilde{S}_1] = 0$. Therefore, action of γ_1 leads to syndrome measurement of $\{-1, +1\}$ when stabilizers $\{S_1, \tilde{S}_1\}$ are measured. Similarly, action of $\tilde{\gamma}_1$ produces syndrome measurement of $\{+1, -1\}$ for stabilizers $\{S_1, \tilde{S}_1\}$. Calculating this for rest of the stabilizer, we get syndrome measurements of $\{-1, +1, -1, +1, -1, +1\}$ due to γ_1 and $\{+1, -1, +1, -1, +1, -1\}$ due to $\tilde{\gamma}_1$ for stabilizers $\{S_1, S_2, S_3, \tilde{S}_1, \tilde{S}_2, \tilde{S}_3\}$. Once the atom loss is detected through syndromes, a new atom is moved/shuttled (from an optical tweezer or a nearby reservoir) into now vacant site and initialized into some known reference state. Depending on the syndrome, correction operator of γ_1 or $\tilde{\gamma}_1$ can be applied. You might wonder what does applying γ_1 or $\tilde{\gamma}_1$ on a fermionic site look like?

γ_k acts the same as X_k gate:

$$\begin{aligned}\gamma_k|0_k\rangle &= (c_k^\dagger + c_k)|0_k\rangle = c_k^\dagger|0_k\rangle + c_k|0_k\rangle = |1_k\rangle \\ \gamma_k|1_k\rangle &= (c_k^\dagger + c_k)|1_k\rangle = c_k^\dagger|1_k\rangle + c_k|1_k\rangle = |0_k\rangle\end{aligned}$$

and $\tilde{\gamma}_k$ acts the same as Y_k gate:

$$\begin{aligned}\tilde{\gamma}_k|0_k\rangle &= i(c_k^\dagger - c_k)|0_k\rangle = ic_k^\dagger|0_k\rangle - ic_k|0_k\rangle = i|1_k\rangle \\ \tilde{\gamma}_k|1_k\rangle &= i(c_k^\dagger - c_k)|1_k\rangle = ic_k^\dagger|1_k\rangle - ic_k|1_k\rangle = -i|0_k\rangle\end{aligned}$$

Both of these are implementable on a digital fermionic quantum device. The table below lists the syndrome signatures for atom loss at each site and the corresponding correction operator:

Atom loss site	Syndrome $(S_1, \tilde{S}_1)$	Syndrome $(S_2, \tilde{S}_2)$	Syndrome $(S_3, \tilde{S}_3)$	Correction
1	$(-1, +1)$ $(+1, -1)$	$(-1, +1)$ $(+1, -1)$	$(-1, +1)$ $(+1, -1)$	γ_1 $\tilde{\gamma}_1$
2	$(-1, +1)$ $(+1, -1)$	$(-1, +1)$ $(+1, -1)$	$(+1, +1)$ $(+1, +1)$	γ_2 $\tilde{\gamma}_2$
3	$(-1, +1)$ $(+1, -1)$	$(+1, +1)$ $(+1, +1)$	$(-1, +1)$ $(+1, -1)$	γ_3 $\tilde{\gamma}_3$
4	$(-1, +1)$ $(+1, -1)$	$(+1, +1)$ $(+1, +1)$	$(+1, +1)$ $(+1, +1)$	γ_4 $\tilde{\gamma}_4$
5	$(+1, +1)$ $(+1, +1)$	$(-1, +1)$ $(+1, -1)$	$(-1, +1)$ $(+1, -1)$	γ_5 $\tilde{\gamma}_5$
6	$(+1, +1)$ $(+1, +1)$	$(-1, +1)$ $(+1, -1)$	$(+1, +1)$ $(+1, +1)$	γ_6 $\tilde{\gamma}_6$
7	$(+1, +1)$ $(+1, +1)$	$(+1, +1)$ $(+1, +1)$	$(-1, +1)$ $(+1, -1)$	γ_7 $\tilde{\gamma}_7$

Finally, we have a fault-tolerant code capable of correcting single-site errors (both phase and loss) on a fermionic device, which will enable us to run our potentially Nobel prize winning experiment!!!

4 Discussion and Open Questions

The landscape of fault-tolerant Majorana fermionic codes has steadily evolved from foundational topological concepts toward high-efficiency quantum architectures. Initial work on Majorana fermion codes (16) established the framework by combining topological protection derived from macroscopic code distance with physical fermionic parity conservation. Building upon this, the Majorana surface code (17) streamlined fault-tolerant measurements by enabling single-step stabilizer checks without requiring physical ancilla qubits. Color codes were later proposed to protect logical Majoranas encoded within physical Majoranas (18), achieving a fault-tolerance error threshold near 0.8%. Expanding these capabilities to modern high-rate paradigms, recent developments in fermion-to-fermion low-density parity-check (LDPC) codes (19) establish viable quantum memory architectures alongside fermionic lattice surgery to facilitate efficient state transfer.

To conclude this lecture, we leave students with the following open questions for further exploration:

- While this lecture assumed a digital fermionic device capable of discrete fermionic gates, how does quantum error correction generalize to analog fermionic processors, which engineer continuous Hamiltonian evolution directly in ultra-cold atomic or optical lattice setups?
- Many of the codes covered today (e.g., Majorana repetition/Steane codes and BKSF similar to toric code) are directly adapted from qubit architectures. Are there genuinely natively fermionic error-correcting codes?
- Is fermionic fault tolerance achievable when employing Majoranas solely as a theoretical framework, or does practical fault tolerance ultimately require physical, topologically protected Majorana modes?

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